COURSE LLM401 / MODULE LL

Embeddings and Vector Search

Learning goal: explain core concepts, compare related methods, and apply them to realistic machine learning or educational technology scenarios.

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01. What is a text embedding?

A text embedding is a numeric vector produced by a model to represent a passage or query. Related meanings often appear close in embedding space. Use the same compatible model and preprocessing for documents and queries. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

02. How do word and sentence embeddings differ?

A static word embedding represents one vocabulary word; a sentence embedding represents a complete text unit. Sentence embeddings suit passage retrieval, while contextual token embeddings vary according to nearby words. A useful way to understand how do word and sentence embeddings differ is to compare the methods not only by definition, but also by the kind of input they expect, the output they produce, and the situations in which each is useful. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

03. How is cosine similarity calculated?

Cosine similarity is the dot product of two vectors divided by the product of their lengths. It compares direction, not magnitude. If vectors are normalized, their dot product equals cosine similarity. In practice, it helps to think of this as a sequence of steps rather than a single operation. Each step affects the information available to the next stage, so implementation choices should be validated on representative data. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

04. How do Euclidean distance and dot product differ?

Euclidean distance measures straight-line separation. Dot product combines angle and vector magnitudes unless vectors are normalized. The index metric should match the embedding model's recommendation and evaluation results. A useful way to understand how do euclidean distance and dot product differ is to compare the methods not only by definition, but also by the kind of input they expect, the output they produce, and the situations in which each is useful. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

05. What does a vector database store?

A vector store indexes embeddings alongside chunk text, document identifiers, and metadata. Examples include FAISS for local similarity indexing and Chroma or managed services for broader storage workflows. Persist model and corpus versions with the index. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

06. What are top-K retrieval and metadata filters?

Top-K retrieval returns the K closest matching candidates. Filters restrict eligible records by fields such as course_id, document_version, or access level. Apply authorization filters before presenting results to a learner. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

07. What does MMR do?

Maximum marginal relevance reranks candidates by balancing similarity to the query against similarity to already selected results. It can reduce redundant chunks and increase diversity, but too much diversity may exclude the strongest evidence. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

08. What is approximate nearest neighbor search?

ANN indexes trade exact exhaustive comparison for lower latency or memory use. Tune the index against held-out queries, measuring recall of relevant neighbors, latency, and storage rather than assuming a faster index is sufficient. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.